Electoral Integrity and Judicial Oversight

GS II

- **Universal adult franchise** allows eligible adult citizens to participate in elections regardless of their social or economic background. Electoral rolls determine who can exercise this right. Errors or unjustified exclusions from these lists can therefore prevent citizens from participating in democratic decision-making.
- Electoral roll revision helps remove incorrect entries and update voter information. However, **verification procedures** must not impose unreasonable difficulties on eligible citizens. When documentation requirements become too demanding, people with limited records may face a greater risk of exclusion.
- The Election Commission is responsible for maintaining electoral integrity, while the Supreme Court can examine whether electoral procedures respect constitutional safeguards. **Judicial oversight** becomes important when citizens challenge administrative decisions that may affect their fundamental democratic participation.
- Effective electoral administration requires accurate records, **transparent procedures** and accessible opportunities to correct mistakes. Independent scrutiny can identify procedural weaknesses, but lasting public confidence also depends on electoral authorities implementing necessary corrections and treating eligible voters fairly.

Electoral integrity requires accurate voter lists and fair verification procedures. Judicial oversight helps protect voting rights when administrative decisions risk excluding eligible citizens.

India's Role in Global Diplomacy

GS II

- International conflicts can affect countries beyond the immediate battlefield through disruptions in **energy supplies**, trade and financial markets. India must therefore consider how wars influence domestic economic stability, particularly when major trading partners or important energy-producing regions are involved.
- **Strategic autonomy** means maintaining the freedom to make foreign-policy decisions according to national interests rather than automatically following another country's position. This allows India to engage with competing powers while evaluating each international issue independently.
- **Diplomatic engagement** provides opportunities to communicate concerns, encourage negotiations and maintain relationships with countries involved in conflict. However, dialogue does not guarantee immediate results because governments may have incompatible security objectives or remain unwilling to compromise.
- India's foreign policy must balance its economic needs, security partnerships and support for peaceful conflict resolution. Sustained communication with different countries can preserve **diplomatic flexibility**, but credibility also depends on consistent principles and practical contributions to reducing tensions.

India's diplomatic approach requires balancing strategic autonomy, economic security and peaceful conflict resolution while maintaining constructive engagement with competing international powers.

Optogenetics and Brain Research

GS III

- The brain contains nerve cells called **neurons** that communicate through electrical and chemical signals. Understanding how particular neurons influence behaviour is difficult because many cells operate simultaneously. Scientists therefore need methods that can selectively activate or silence specific groups of neurons.
- Optogenetics combines genetic techniques with light-based stimulation. Scientists introduce genes that enable selected cells to produce **light-sensitive proteins**. When exposed to suitable wavelengths of light, these proteins can change electrical activity in the targeted cells.
- By controlling selected neurons, researchers can investigate how **brain circuits** contribute to movement, memory, emotions and other functions. Observing changes after activating or suppressing particular cells helps scientists identify relationships between neural activity and behaviour.
- Optogenetics offers opportunities to study neurological and psychiatric disorders by identifying how particular brain circuits function. However, translating laboratory findings into **medical treatments** requires further research because delivering light and targeting cells safely in humans presents technical challenges.

Optogenetics uses light-sensitive proteins to control selected neurons, helping researchers investigate brain circuits and develop a deeper understanding of neurological functions and disorders.

Great Indian Bustard Conservation

GS III

- The Great Indian Bustard is a critically endangered bird that depends on **open grasslands** and semi-arid landscapes. Its survival requires suitable breeding grounds, food availability and protection from disturbance. Conservation therefore involves maintaining the ecological conditions necessary for the species to reproduce.
- Grassland habitats face pressure from agricultural expansion, infrastructure development and **changing land use**. Although tree planting is often associated with environmental improvement, converting natural grasslands into wooded areas can harm species adapted to open landscapes.
- **Overhead power lines** create serious risks for large flying birds such as the Great Indian Bustard. Collisions can cause deaths even when surrounding habitats remain suitable. Measures such as underground cabling in priority areas and appropriate bird-diversion devices can reduce these threats.
- **Captive breeding** can help increase the population of a threatened species, but releasing birds without protecting their natural habitats may not produce lasting recovery. Successful conservation requires scientific monitoring, habitat restoration, cooperation with local communities and action against continuing environmental threats.

Great Indian Bustard conservation requires protecting natural grasslands, reducing infrastructure-related mortality and combining captive breeding with long-term habitat management.